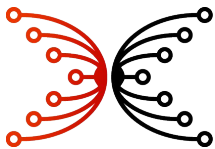


Input | Output Group



Cyclic Atomic N-party Swap (CANS) Protocol

Trustless Cross-chain Swaps via Adaptor Signatures

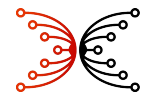


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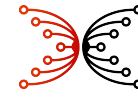
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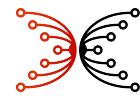
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Motivations

Why move assets between blockchains?

- Untapped liquidity: Bitcoin has largest store of value in crypto, approx \$1 trillion in market cap, not earning interest.
- Bitcoin scripting is intentionally limited for security reasons, so no native lending, no DEX, no stable-coins.
- Cardano has rich programmable finance: oracles, DEXs, lending & borrowing, liquid staking – but not Bitcoin's liquidity.
- If you sell one for the other, lose exposure and potentially taxable event.

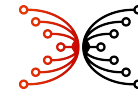


Motivations

Risk of Bridges

- Bridges use wrapped tokens (lets you keep exposure) that you can then yield.
- Most bridges require a trusted third-party/custodian to upgrade the contract logic, pause or drain the bridge.
- Fully-custodial, multi-sig validators, optimistic bridges, zk-bridges.
- Bridges have lost >\$2B to exploits.

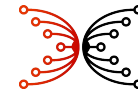
Bridge	Date	Amount	Chains Affected	Exploit Type	Backstop?
Poly Network	Aug 2021	\$600M	Ethereum, BSC, Polygon	Smart contract logic flaw	Yes — hacker returned funds
Wormhole	Feb 2022	\$320M	Solana → Ethereum	Signature verification bug, wETH minted from nothing	Yes — Jump Trading replaced funds
Ronin	Mar 2022	\$625M	Ronin sidechain → Ethereum	5/9 validator keys compromised	No
Nomad	Aug 2022	\$190M	Ethereum, Moonbeam, Avalanche, Evmos	Botched upgrade, anyone could spoof messages	No
Orbit Chain	Jan 2024	\$81M	Ethereum → multiple	7 of 10 multisig keys compromised Investing.com	No
Multichain	Jul 2023	\$126M	Fantom (largest at \$118M), Ethereum, Avalanche, BSC MEXC	CEO arrested in China, keys confiscated — possible inside job CoinMarketCap	No — protocol shut down
Kelp DAO	Apr 2026	\$292M	20 chains via LayerZero CoinLore	Bridge drain	No



Motivations

Risks of Hashed Timelock Contracts

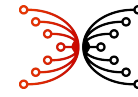
- CEXs impose custody risk, KYC requirements, and settlement delays
- Atomic swaps require no custodian, no shared pool. You own the asset
- HTLC-based atomic swaps publish a common hash on both chains — trivially linking both swap legs for chain analysts
- HTLC ring swaps lock sequentially — each hop waits for the previous confirmation, making total lock-time $O(N \times \text{block_time})$; timeouts must be staggered by the same factor
- Goal: Atomic swap that is efficient, trustless, private, and has minimal on-chain footprint



Solution

Cyclic Atomic N-party Swap (CANS) Protocol Summary

- **CYCLIC**: $P_1 \rightarrow P_2 \rightarrow \dots \rightarrow P_n \rightarrow P_1$: each participant sends exactly once and receives exactly once
- **ATOMIC**: Either all spend txs complete, or all refunds fire. If any participant aborts, time-locked refunds guarantee no participant loses funds
- **N-party**: Scales to N participants – same ring protocol handles 3 or 20 counterparties with no structural changes
- Supports same-chain or cross-chain legs (BTC $\leftrightarrow$ ADA, ADA $\leftrightarrow$ ADA, BTC $\leftrightarrow$ BTC)
- Participants need only trust the cryptographic protocol – not each other
- Scriptless on Bitcoin: Taproot key-path spend is indistinguishable from a regular transfer
- Happy path is quick (refund is no quicker)



Solution

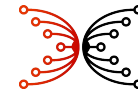
Adaptor Schemes (Secret Sauce)

An adaptor signature is a pre-signature that becomes valid when a secret scalar is added,

i.e. **valid_sig = pre_sig + secret**

- Adaptor point is public commitment to adaptor secret and is shared early.
- The adaptor point alone is enough to create and verify a pre-signature — the secret is only needed at claim-time to complete it into a valid signature.
- Unlike hashed timelocks, the mathematics of adaptor points allows secrets to be combined — making the multi-party sum construction possible.
- Each participant holds one adaptor secret per swap leg; the secret is only revealed at claim-time.
- Collective secret for swap is sum of individual secrets (i.e. $t_1 + \dots + t_n$).
- Publishing a valid spend Tx reveals the secret to all chain observers.

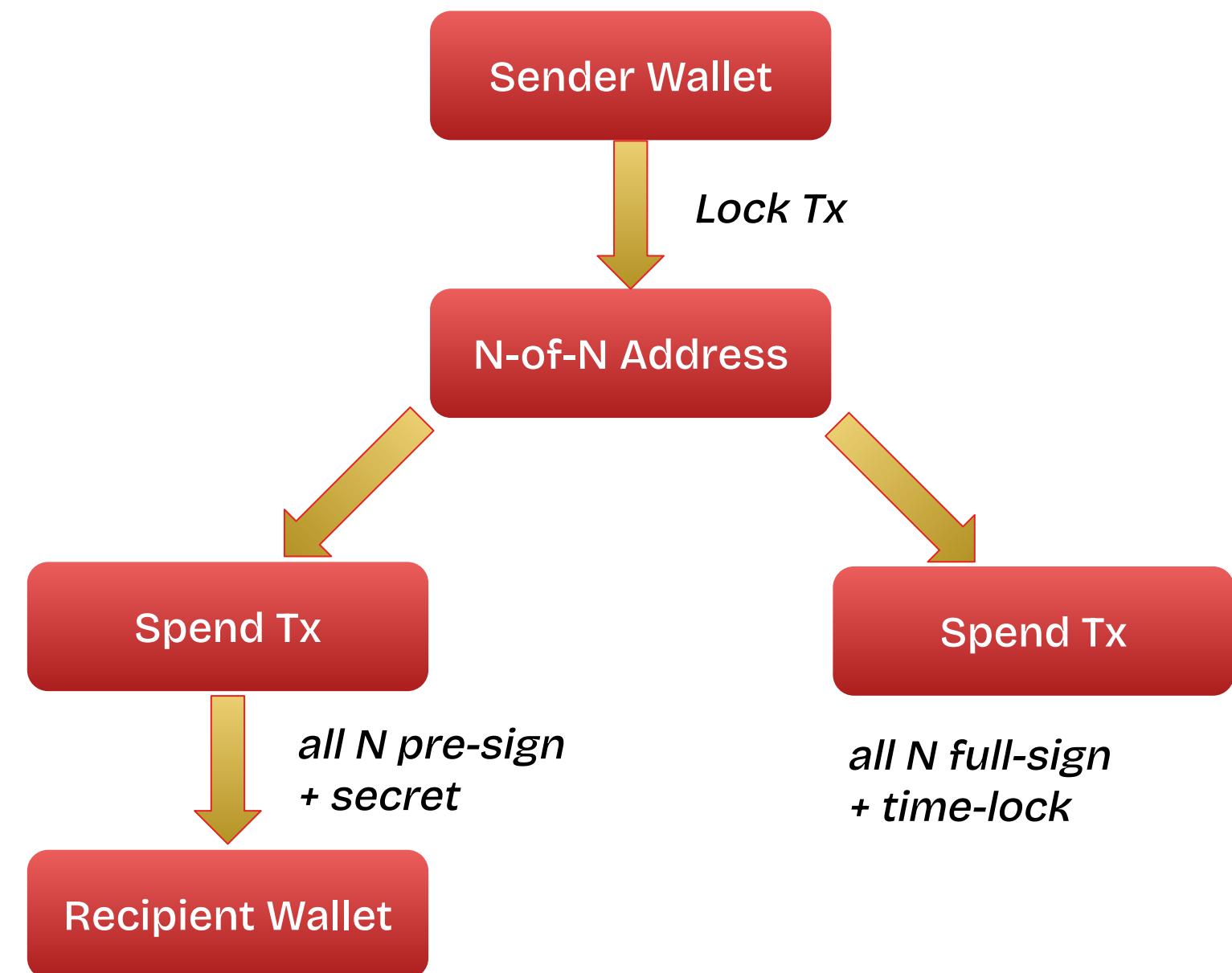


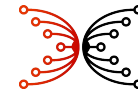


Solution

Lock, Spend and Refund Transactions

- Three Txs prepared per leg before any lock is broadcast:
 - Lock Tx — signed by sender only; funds sent to N-of-N aggregate address (Taproot / Plutus);
 - Spend Tx — pre-signed by all N participants; adaptor secret needed to complete ($t_1 + t_2 + \dots + t_n$);
 - Refund Tx — fully signed by all N participants; valid only after specific block/slot, if swap fails.
- Refund Txs are fully signed and spend Txs are aggregated into adaptor signatures before any lock Tx is broadcast — no party locks funds without a guaranteed exit path and a spend tx that needs only the adaptor secret to finalise.

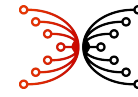




Solution

Bitcoin and Cardano

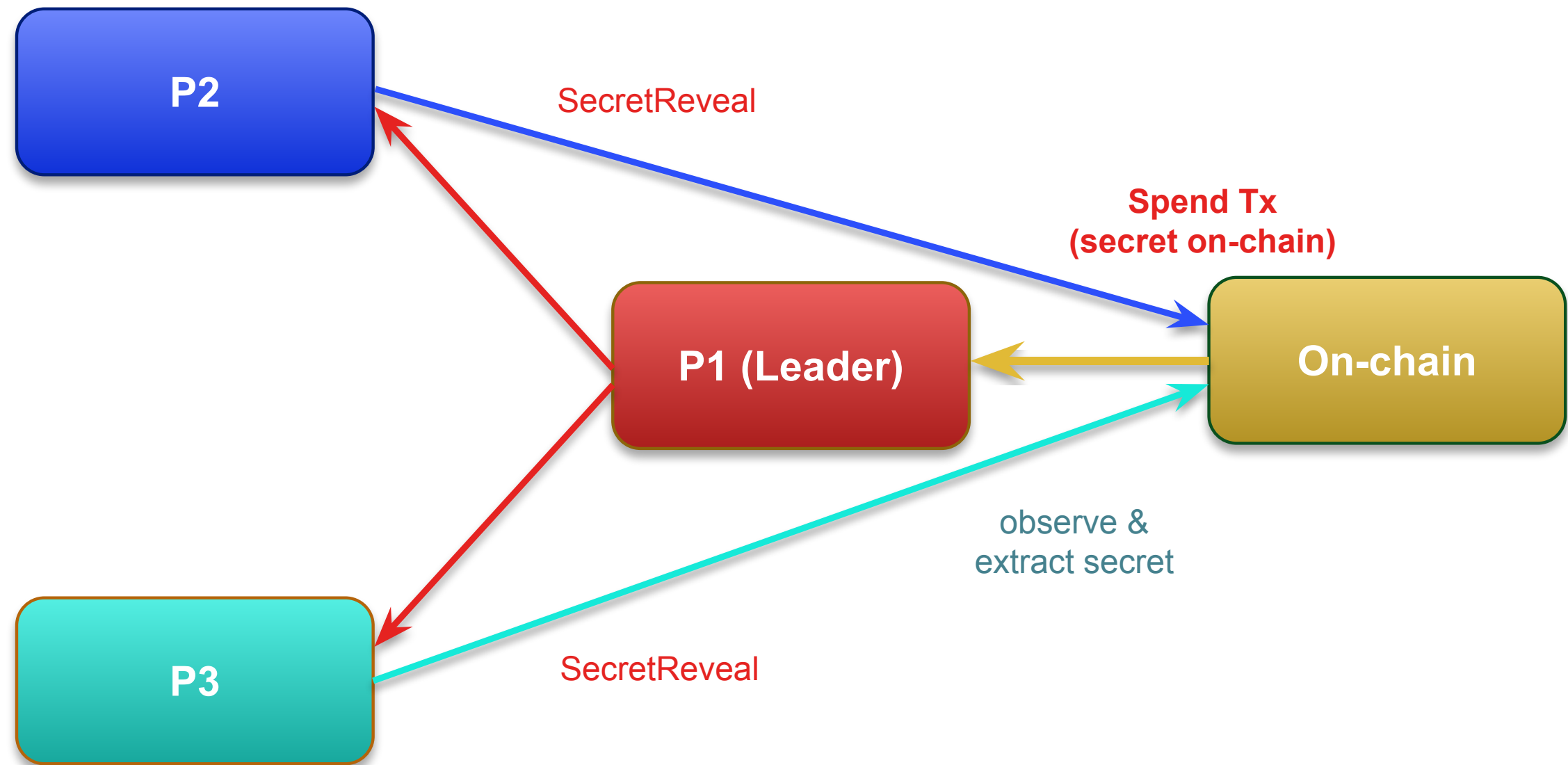
- Bitcoin — Taproot / MuSig2
 - Pay-to-Taproot key-path spend: looks like a regular single-sig transaction on-chain
 - N-of-N key aggregation via MuSig2 — private, efficient, no script visible
 - Adaptor signatures integrate naturally into the Schnorr two-round signing protocol
- Cardano — Plutus
 - Cardano doesn't natively support Schnorr (EdDSA)
 - Locked UTXO guarded by a Plutus validator script
 - Script checks: correct N-of-N multisig
 - Supports any native Cardano asset — not limited to ADA

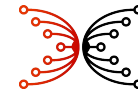


Solution

Leader Election and Secret Sharing

- Leader elected via commit-reveal (hash of random nonce)
- Non-leaders each send their adaptor secret to the leader (once Lock Tx is broadcast).
- Leader adapts and broadcasts Spend Tx – secret becomes visible on-chain.
- Non-leaders observe the Spend Tx and extract the secret to claim their funds.

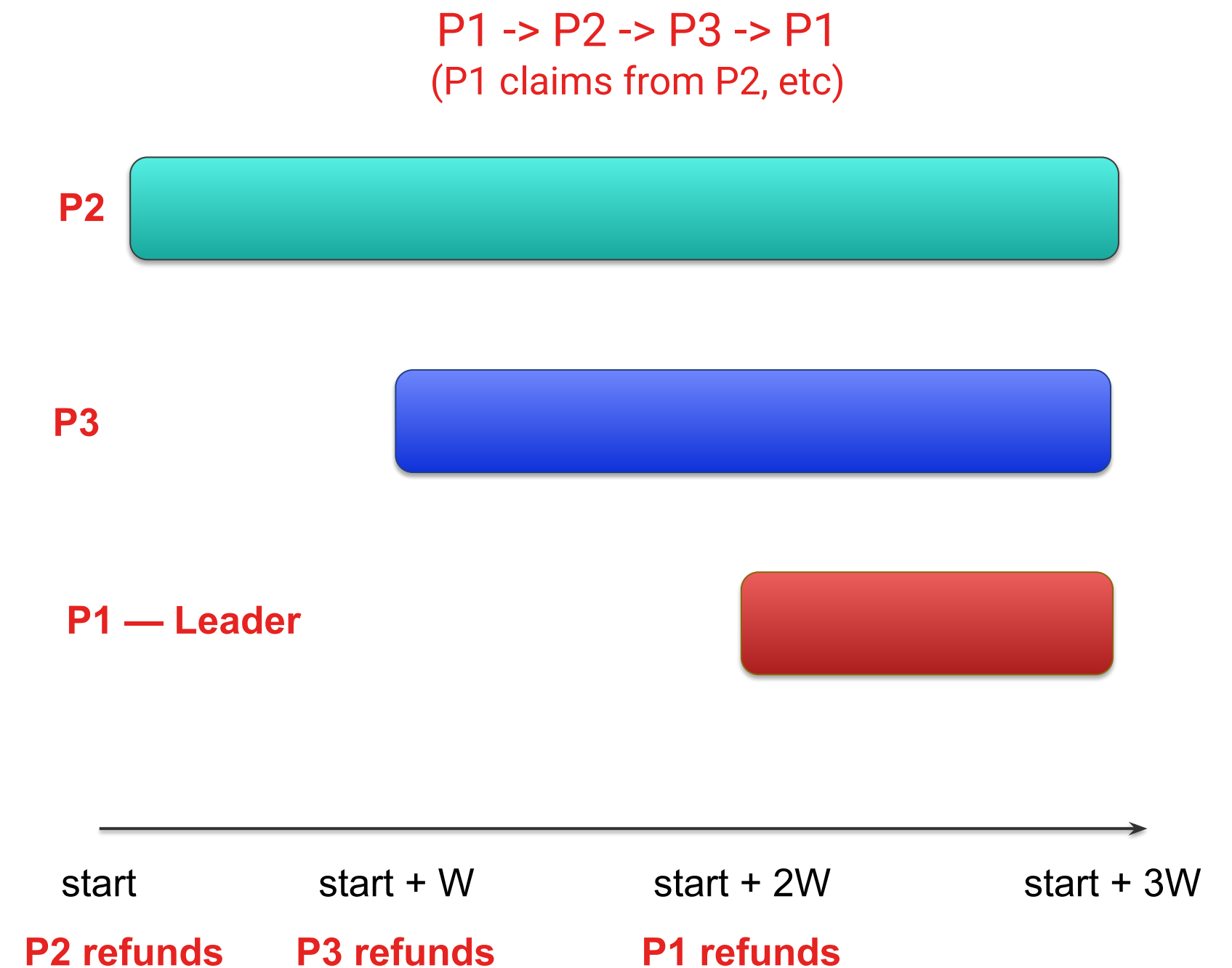


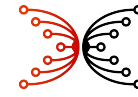


Solution

Refund Windows

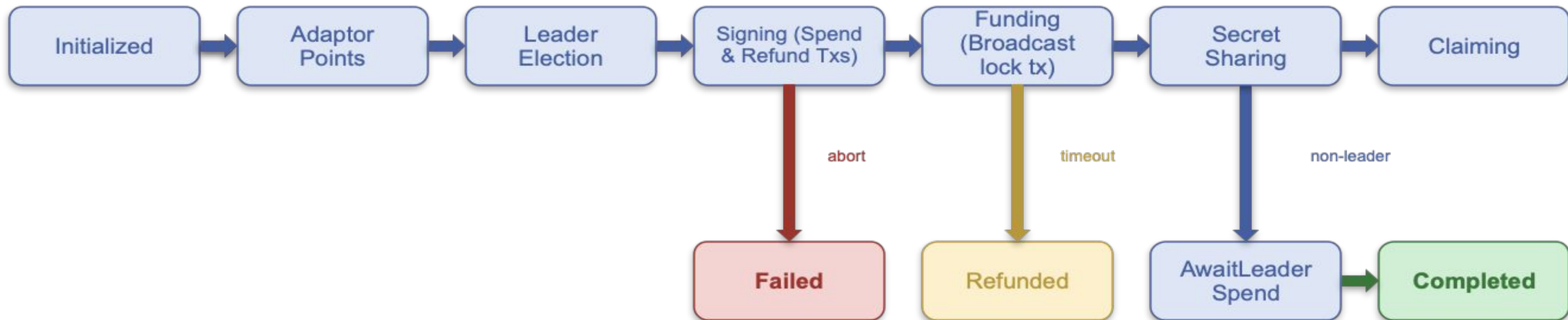
- A refund tx is fully-signed but only valid after a certain block/slot height.
- The participant whose refund-window opens first is the one the leader claims from to create economic pressure to broadcast the spend tx promptly.
- A participant's refund window opens after the window of whoever is claiming from them.
- The leader waits longest — they cannot reclaim until all others have had the chance to exit.
- Guarantees no participant can simultaneously claim their deposit and claim their target's funds.
- Total refund horizon grows linearly with ring size.





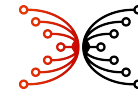
Code

Session States



Leader path: Initialized → AdaptorPts → LeaderElect → Signing → Funding → SecretSharing → Claiming → Completed

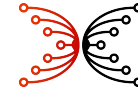
Non-leader fork: SecretSharing → AwaitLeaderSpend → Claiming → Completed



Code

Wire Messages

- Peer-to-peer TCP messages drive state transitions across all daemons:
 - AdaptorPoint – initial secret commitment broadcast to all peers,
 - LeaderElectionCommitment / LeaderElectionNonce – commit-reveal for leader selection,
 - SchnorrNonce / PartialSignature – MuSig2 two-round signing,
 - LockTxBroadcast – participant signals their lock tx is on-chain,
 - SecretReveal – each participant sends adaptor secret to leader,
 - SpendTxBroadcast – leader notifies others the spend tx is live.
- No central coordinator – every message is routed directly between peers.



Code

Architecture

- Each participant runs an independent [daemon process](#)
- Components:
 - TCP listener — peer-to-peer, no central server,
 - Chain pollers — watches Bitcoin/Cardano for lock confirmations, leader spend txs, refund windows,
 - Session Finite State Machine — routes incoming messages into the SwapSession
 - Key management — BTC Taproot keys, ADA Ed25519 / Plutus keys
- OpenTelemetry tracing for distributed observability across all daemons
- Written in Rust — memory-safe, async (Tokio), tested via a full regtest harness with UI dashboard
- [Architectural documentation](#) maps between the protocol specification and the reference implementation.



Demo

20-Party Swap

Video Length 2':12"

Colour Code

Initialized

Awaiting Adaptor Points

Awaiting Leader Commitments

Awaiting Leader Nonces

Refund Txn Signing

Funding

Awaiting Lock Confirmations

Awaiting Secrets

Awaiting Leader Spend

Claiming

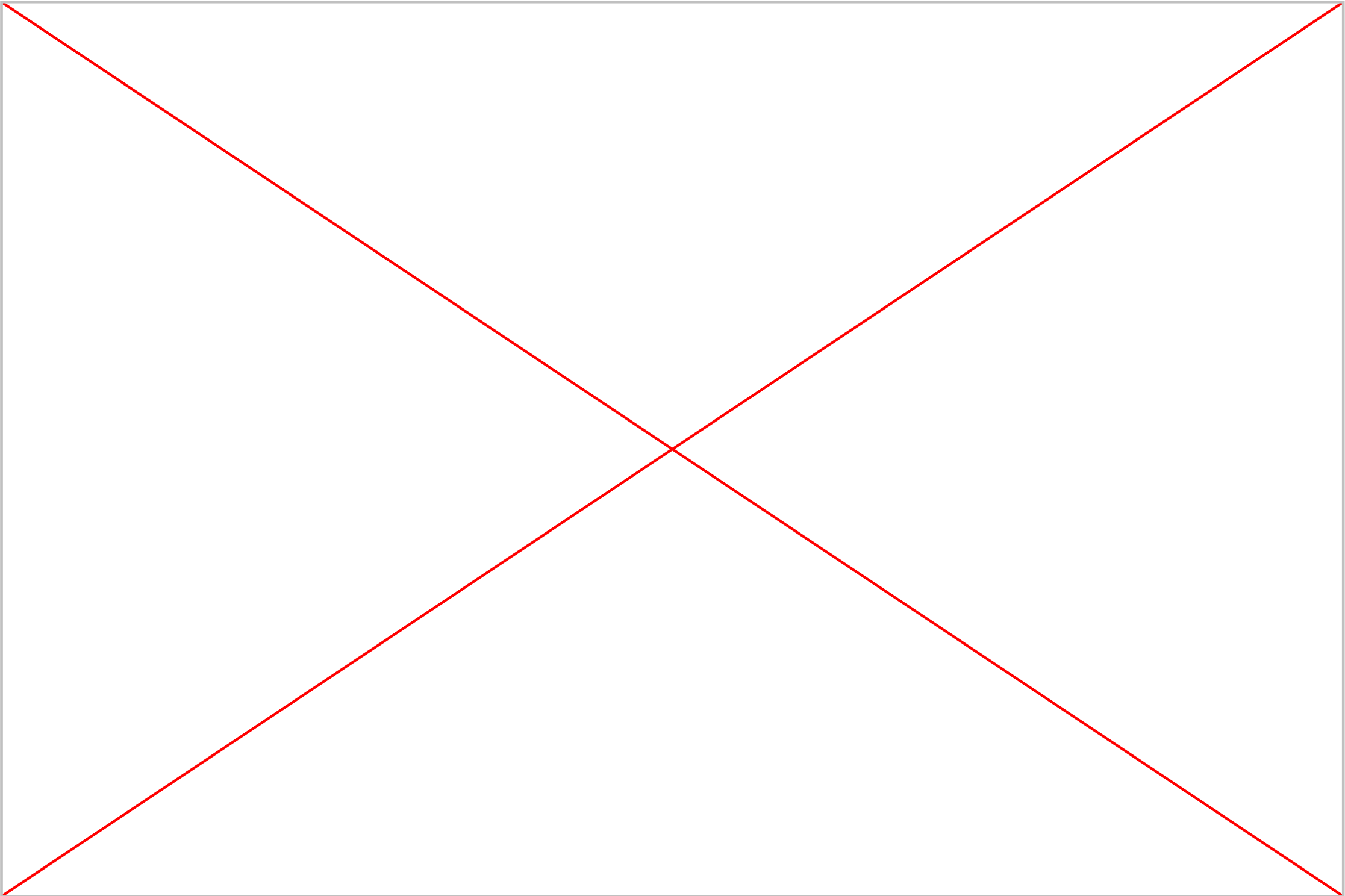
Completed

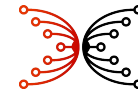
Refunded

Failed

Bitcoin participant

Cardano participant

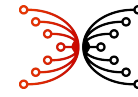




Challenges

Open Issues

- Ring-Structure
 - Participants must form a connected ring.
 - Even non-overlapping rings are forbidden from occurring in the same swap.
- DoS Attack
 - A censoring miner suppresses a participant's spend tx until the target's refund window opens, at which point the target reclaims their funds and the spend tx becomes invalid.
- Cross-chain clock coordination
 - Bitcoin and Cardano advance independently — no universal block clock.
 - Refund windows must be calibrated conservatively to avoid timing races.
- Refund window length scales with N
 - Outermost participant waits up to $N \times \text{window_duration}$ before a safe refund.
 - Practical upper bound on ring-size for time-sensitive swaps.



Thank you for listening

Questions?

<https://discord.com/invite/56EQWW4rzv>

<https://www.iog.io/contact>